



CROSS-EYED

An Arabic drawing, nearly 1,000 years old, shows the optic nerves crossing. Half of the nerve fibers from the right eye actually pass to the left side of the brain, and vice versa.

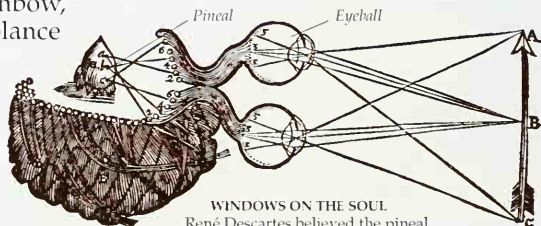
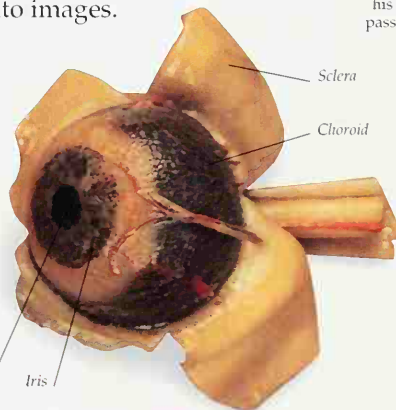
Eyes and seeing

THE BODY ACTUALLY HAS MORE than just five senses of sight, hearing, smell, taste, and touch. Sensors around the body can detect temperature, positions of body parts such as muscles and joints, and levels of oxygen, nutrients, and other chemicals in body fluids. The main sense is sight. Two-thirds of the mind's conscious attention is taken up by what the eyes see, and two-thirds of information stored in the brain has come in by vision – as pictures, words, and other visual stimuli. At Alexandria, Egypt, in the first century AD, the anatomist Rufus of Ephesus described the main parts of the eye: the domelike cornea at the front, the colored iris or “rainbow,” the lens (named from its resemblance

to a lentil), and the vitreous humor, or “glassy fluid” inside the eyeball. Only with the microscope (pp. 12-13) could anatomists see the millions of rod and cone cells in the retina, which is the thin layer on the inside of the eyeball. These cells detect light rays, turn them into nerve signals, which the optic nerve transmits to the brain for interpretation into images.

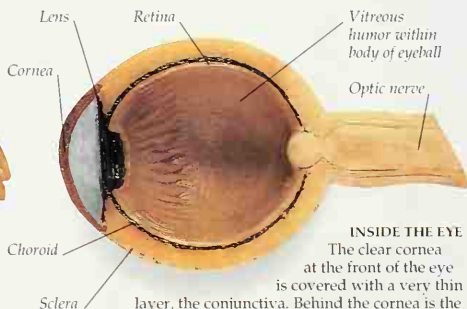
OUTSIDE THE EYE

The wall of the eyeball is a three-layered sandwich. Outermost is the sclera, pale and tough, and visible at the front as the “white” of the eye. In the middle is the choroid, dark and spongy, and rich in blood vessels. The innermost layer is the light-detecting retina (“net”). Tissue-thin, with a working area hardly bigger than a thumbnail, it detects an incredibly detailed, full-color view of the world.



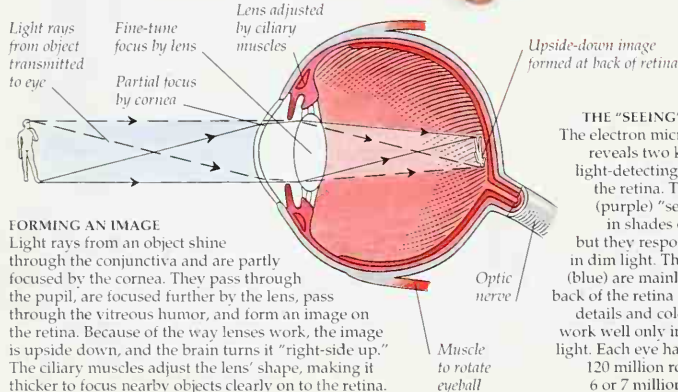
WINDOWS ON THE SOUL

René Descartes believed the pineal, a pea-sized gland in the brain (pp. 60-61), was the seat of the soul. This drawing from *Principia Philosophiae* (1644) shows his mistaken theory that “animal spirit” carrying visual information passed from the eyes along hollow optic nerves, directly to the pineal.



INSIDE THE EYE

The clear cornea at the front of the eye is covered with a very thin layer, the conjunctiva. Behind the cornea is the iris, a colored ring of muscles around a hole in its center, the pupil. The ring automatically widens in bright light to reduce the size of the pupil, thus protecting the eye's delicate retina from too much potentially damaging light.



FORMING AN IMAGE

Light rays from an object shine through the conjunctiva and are partly focused by the cornea. They pass through the pupil, are focused further by the lens, pass through the vitreous humor, and form an image on the retina. Because of the way lenses work, the image is upside down, and the brain turns it “right-side up.” The ciliary muscles adjust the lens' shape, making it thicker to focus nearby objects clearly on to the retina.

THE “SEEING” CELLS
The electron microscope reveals two kinds of light-detecting cells in the retina. The rods (purple) “see” only in shades of gray, but they respond well in dim light. The cones (blue) are mainly at the back of the retina and see details and colors, but work well only in bright light. Each eye has about 120 million rods and 6 or 7 million cones.

